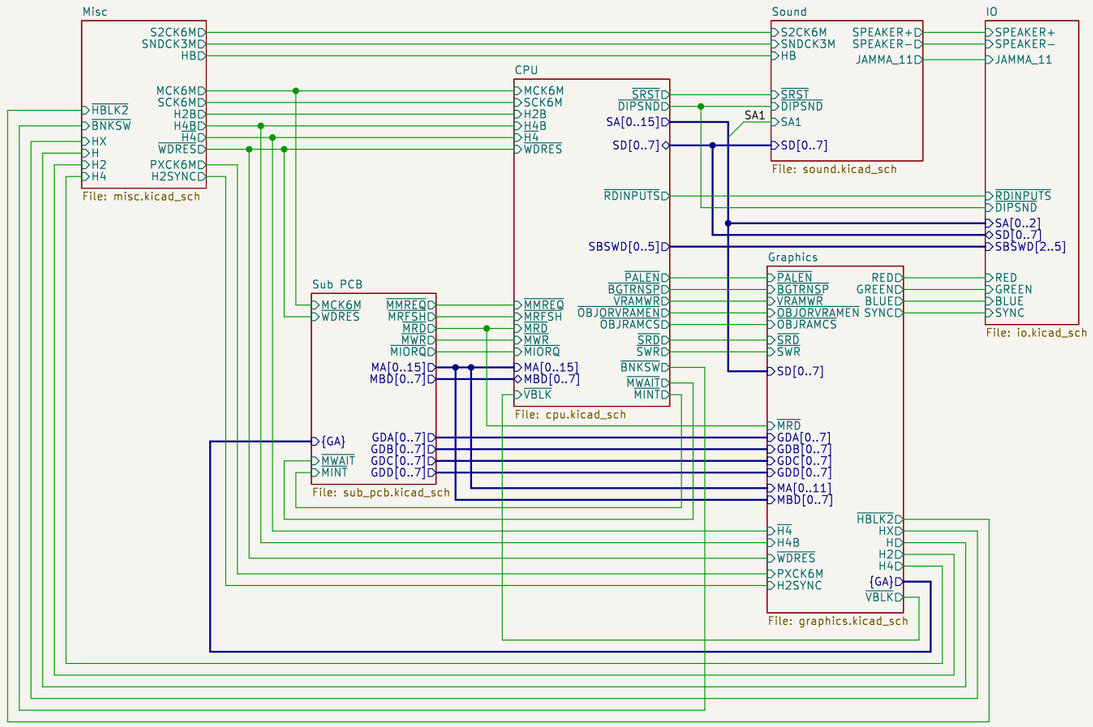


Note: The reference designator refers to the position on the PCB. If the reference designator already exists the letter is doubled. I.e. C1 -> CC1.

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Sheet: /		
File: tnzs.kicad_sch		
Title: The NewZealand Story – P0-043A		
Size: A3	Date: 2024-04-07	**Rev:**
KiCad E.D.A. 8.0.5		Id: 1/7

IC5
74LS174
MD0 MD1 MD2 MD3 MD4 MD5
D0 D1 D2 D3 D4 D5
Q0 Q1 Q2 Q3 Q4 Q5
Cp
Mr
hex flip flop
VCC
GND

IC1
74F245
MD0 MD1 MD2 MD3 MD4 MD5 MD6 MD7
A0 A1 A2 A3 A4 A5 A6 A7
B0 B1 B2 B3 B4 B5 B6 B7
buffers
A->B
CE
MRD
VCC
GND
MBD[0..7]
MBD[0..7]

MA11 MA12 MA13 MA14 MA15
MCK6M
MBD4 MBD3 MBD5 MBD6
MBD2 MBD7 MBD0 MBD1
MINT
MBUSRQ
MHALT
MMREQ
MIORQ

JS2
Conn_01x20
VCC
GND

MA10 MA9 MA8 MA7 MA6 MA5 MA4 MA3 MA2 MA1 MA0
MRFSH
MM1
WDRES
MBUSRQ
MWAIT
MBUSACK
MWR
MRD

JS1
Conn_01x20

IC6
PAL16L8'
MA10 MA11 MA12 MA13 MA14 MA15
GROMENB
MWR
MRD
MMREQ
I1 I2 I3 I4 I5 I6 I7 I8 I9 I11
I/O13 I/O14 I/O15 I/O16 I/O17 I/O18 O12 O19
GROMENB MDC2 MDC1 MDC0 MA9 MRAMCS
VCC
VSS
GND

/MRAMCS = /MA14 & MA15 & /MDC2 & /MDC1
o13 = /MA14 & MA15 & /MDC2 & /MDC1
/GROMENB = GROMEN
/o19 = MA9 & MA10 & /MA11 & MA12 & MA14 & MA15 & /MWR & /MMREQ

MA[0..15]
MA[0..15]

IC3
Z80CPU
WDRES
IC4C
MCK6M
74HC08
MINT
MM1 MRFSH MWAIT MHALT
MRD MWR MMREQ MIORQ
MBUSRQ MBUSACK
RESET CLK NMI INT M1 RFSH WAIT HALT RD WR MREQ IORQ BUSRQ BUSACK
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15
D0 D1 D2 D3 D4 D5 D6 D7
MA0 MA1 MA2 MA3 MA4 MA5 MA6 MA7 MA8 MA9 MA10 MA11 MA12 MA13 MA14 MA15
MD0 MD1 MD2 MD3 MD4 MD5 MD6 MD7
MIORQ MMREQ MWR MRD MRFSH
VCC
GND

IC2
SRM20256
MA0 MA1 MA2 MA3 MA4 MA5 MA6 MA7 MA8 MA9 MA10 MA11 MA12 MA13 MDC0
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14
Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7
MD0 MD1 MD2 MD3 MD4 MD5 MD6 MD7
CS
OE
WE
MRD
MWR
VCC
GND

MD[0..7]

IC7 IC8 IC9 IC10 IC11 IC13 IC12 IC14
27C010
GA0 GA1 GA2 H8R GA4 GA5 GA6 GA7 GA8 GA9 GA10 GA11 GA12 GA13 GA14 GA15 GA16
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15 A16
D0 D1 D2 D3 D4 D5 D6 D7
GDA0 GDA1 GDA2 GDA3 GDA4 GDA5 GDA6 GDA7
GDB0 GDB1 GDB2 GDB3 GDB4 GDB5 GDB6 GDB7
GDC0 GDC1 GDC2 GDC3 GDC4 GDC5 GDC6 GDC7
GDD0 GDD1 GDD2 GDD3 GDD4 GDD5 GDD6 GDD7
VCC
PGM
VPP
CE
OE
GND
GROMEN
GROMENB

NC: MM1, MBUSRQ, MHALT, MBUSACK

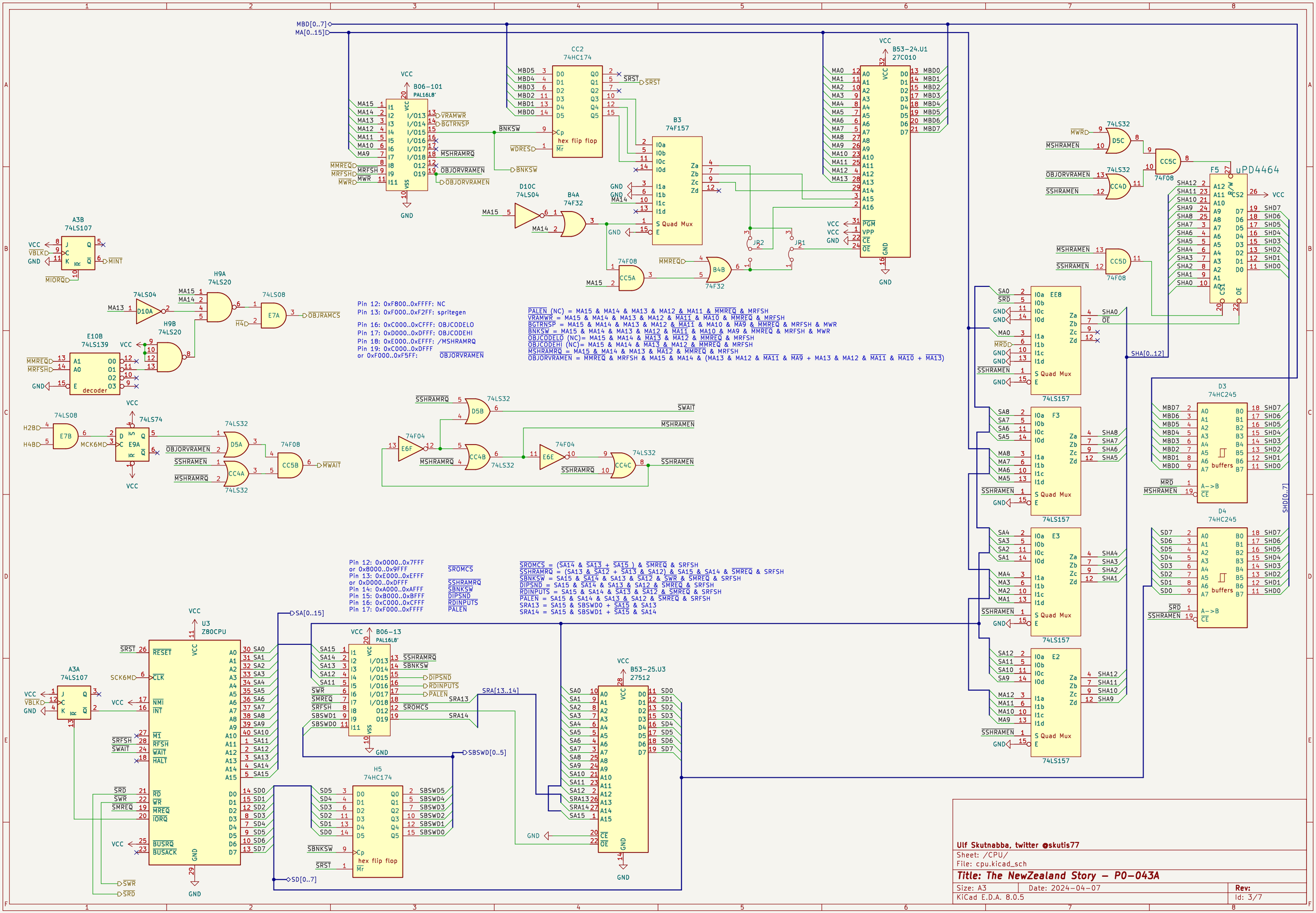
{GA}

JS6 JS5 JS4 JS3 JS10 JS9 JS8 JS7
Conn_01x20
GND
VCC
GDB0 GDB1 GDB2 GDB3 GDB4 GDB5 GDB6 GDB7
GDA0 GDA1 GDA2 GDA3 GDA4 GDA5 GDA6 GDA7
GA8 GA7 GA6 GA5 GA4 H8R GA2 GA1 GA9 GA10 GA11 GA12 GA13 GA14 GA15 GA16 GROMEN GA0
GDD0 GDD1 GDD2 GDD3 GDD4 GDD5 GDD6 GDD7
GDC0 GDC1 GDC2 GDC3 GDC4 GDC5 GDC6 GDC7

GDA[0..7]
GDB[0..7]
GDD[0..7]
GDC[0..7]

IC4D
74HC08
IC4B
74HC08
IC4A
74HC08
GND

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Sheet: /Sub PCB/
File: sub_pcb.kicad_sch
Title: The NewZealand Story – P0-043A
Size: A3 | Date: 2024-04-07 | Rev:
KiCad E.D.A. 8.0.5 | Id: 2/7



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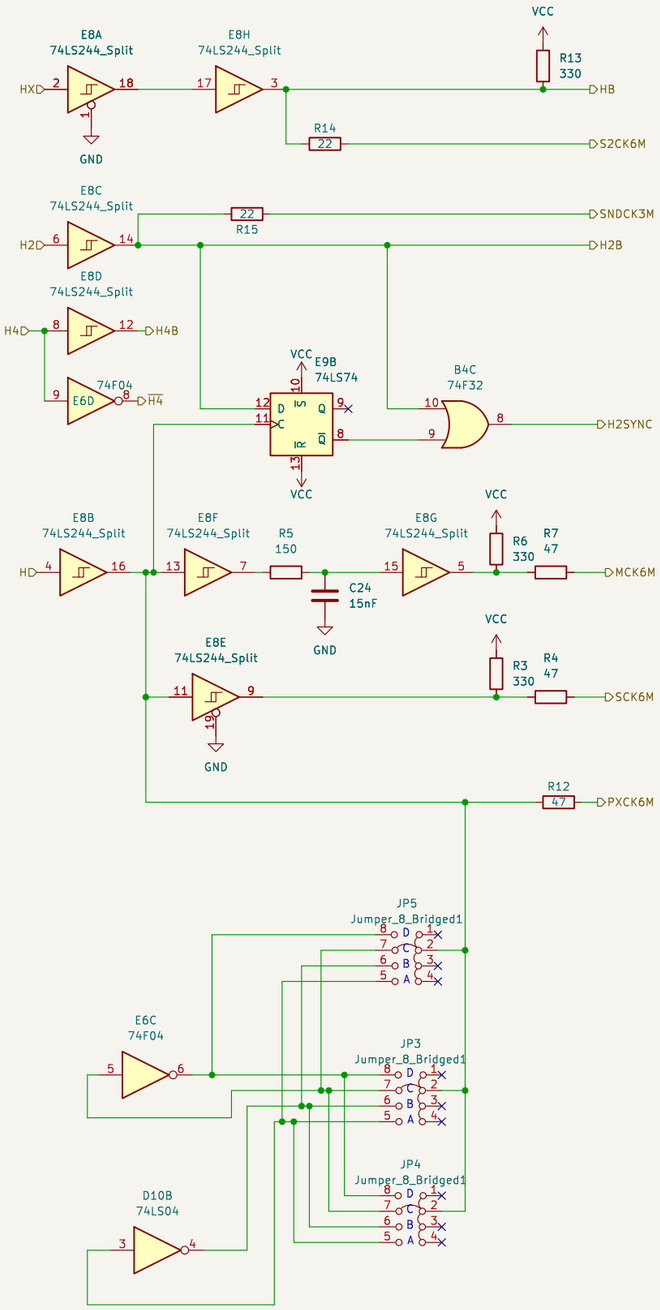
Sheet: /CPU/

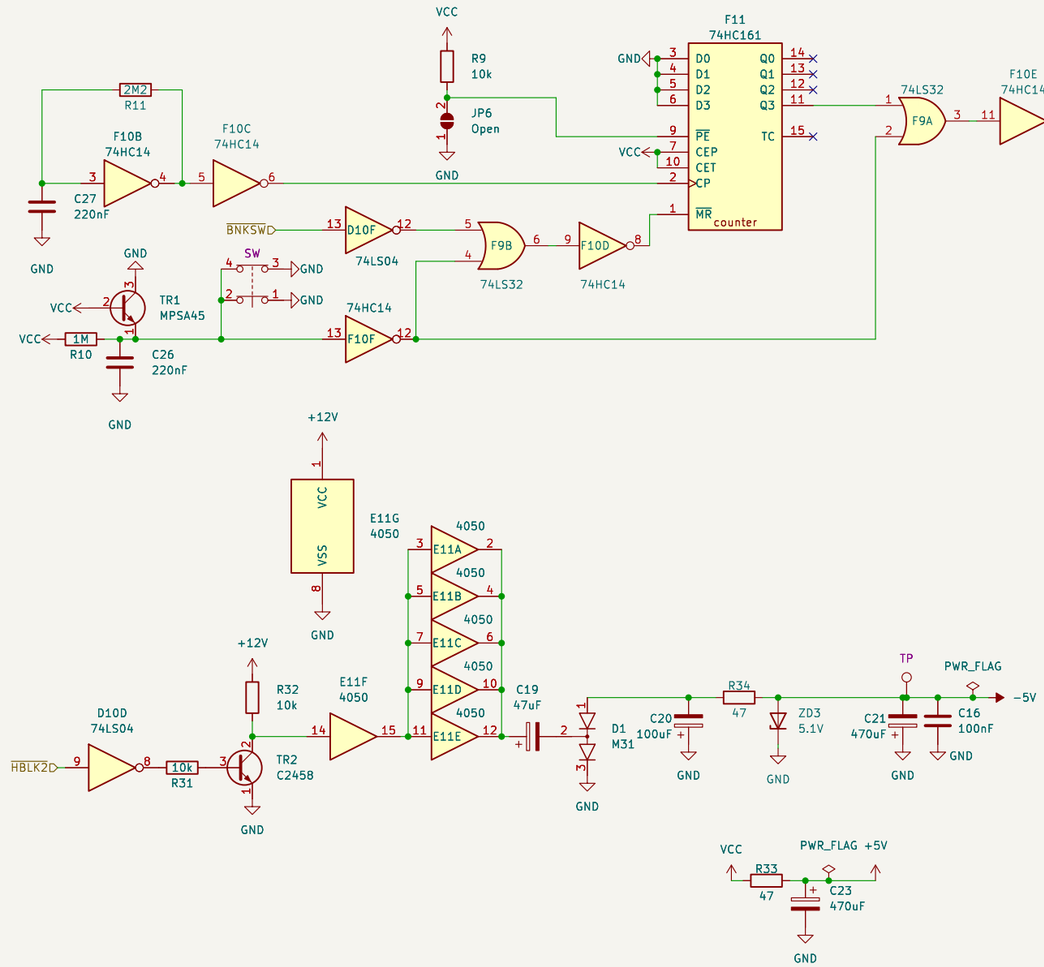
File: cpu.kicad_sch

Title: The NewZealand Story – P0-043A

Size: A3	Date: 2024-04-07	Rev:
KiCad E.D.A. 8.0.5		Id: 3/7

CC3
74HC245
MBD[0..7]
MBD7
MBD6
MBD5
MBD4
MBD3
MBD2
MBD1
MBD0
A0
A1
A2
A3
A4
A5
A6
A7
buffers
A->B
CE
B0
B1
B2
B3
B4
B5
B6
B7
VD7
VD6
VD5
VD4
VD3
VD2
VD1
VD0
D5D
74LS32
H4B
OBJORVRAMEN
MRD
D6
74HC374
D0
D1
D2
D3
D4
D5
D6
D7
O0
O1
O2
O3
O4
O5
O6
O7
3-state
flip flops
H2SYNC
H4
Cp
OE
VDB7
VDB6
VDB5
VDB4
VDB3
VDB2
VDB1
VDB0
VDB[0..7]
CC6
OSC-12.0000MHz
VCC
GND
OUT
NC
R1
330
R2
22
D8
YM3906
CLK0
12M
VBLK
HBLK2
VSYNC
HBLK
DATAEN
HC8
HC7
HC6
HC5
HC4
HC3
HC2
HC1
HC0
H8
H4
H2
H
HX
{HCNT}
CHA0
CHA1
CHA2
CHA4
GA0
GA1
GA2
GA4
H8B
H8R
SPBANK2
SPBANK1
MA11
-{WDRES
WDRES
A11
VA11
OBJRD
VA11
OBJRD
UK
VSS
SETA_X1-001
VRAMWR
FLIPX
FLIPY
XO0
VA[0..12]
MRD
E6A
74F04
OBJRD
B4D
74F32
E6B
H2SYNC
H8
74LS253
MA12
I0a
I1a
I2a
I3a
4-to-1 mux
OEa
Za
I0b
I1b
I2b
I3b
OEb
Zb
A0
A1
GND
A5
6264SL
VA12
VA11
VA10
VA9
VA8
VA7
VA6
VA5
VA4
VA3
VA2
VA1
VA0
R/W
CS2
CS1
OE
OBJRAMCS
74HC245
MA5
MA4
MA3
MA2
MA1
MA0
A2 and B2 use a modified symbol
with A inputs fixed as inputs
in order to avoid DRC errors.
MA[0..11]
Fixed with VCC
MAB11
74HC273
74HC174
GA12
GA11
GA10
GA9
GA8
GA7
GA6
GA5
GROMEN
GA16
GA15
GA14
GA13
flip flops
hex flip flop
Cp
Mr
VD[0..7]
{GA}
CC8
YM3909
BGTRNSP
CLK
PAL4/TRSP
PAL3
PAL2
PAL1
PAL0
UK
DATAEN
HC8
HC7
HC6
HC5
HC4
HC3
HC2
HC1
HC0
GDA0
GDA1
GDA2
GDA3
GDA4
GDA5
GDA6
GDA7
GDB0
GDB1
GDB2
GDB3
GDB4
GDB5
GDB6
GDB7
GDC0
GDC1
GDC2
GDC3
GDC4
GDC5
GDC6
GDC7
GDD0
GDD1
GDD2
GDD3
GDD4
GDD5
GDD6
GDD7
SETA_X1-002
GDD[0..7]
GDC[0..7]
GDB[0..7]
GDA[0..7]
From sub PCB connectors
To sub PCB connectors
B11
X1-006
SRD
SWRD
PALEN
RD
WR
SD[0..7]
SD0
SD1
SD2
SD3
SD4
SD5
SD6
SD7
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
COL0
COL1
COL2
COL3
COL4
COL5
COL6
COL7
COL8
SA1
SA2
SA3
SA4
SA5
SA6
SA7
SA8
SA9
SA0
SA[0..9]
BB12
X1-007
RGB1
RGB2
RGB3
RGB4
RGB5
RGB6
RGB7
RGB8
RGB9
RGB10
RGB11
RGB12
RGB13
RGB14
RGB15
PXCK6MD
PXCLK
VBLK
VSYNC
HSYNC
CSYNC
R[1..6]
G[1..6]
B[1..6]
R0-012
RED
GREEN
BLUE
R8
47
SYNC
C25
150nF
EE10
74HC161
F10A
74HC14
counter
PE
CEP
CET
TC
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Sheet: /Graphics/
File: graphics.kicad_sch
Title: The NewZealand Story – P0-043A
Size: A3
Date: 2024-04-07
KiCad E.D.A. 8.0.5
Rev:
Id: 4/7





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Sheet: /Misc/
File: misc.kicad_sch

Title: The NewZealand Story – P0-043A

Size: A3	Date: 2024-04-07	**Rev:**
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RA3
2.2K x 8
1P_START 2
1P_BTN_3 3
1P_BTN_2 4
1P_BTN_1 5
1P_DOWN 6
1P_UP 7
1P_RIGHT 8
1P_LEFT 9
1 VCC

RA4
2.2K x 8
2P_START 2
2P_BTN_3 3
2P_BTN_2 4
2P_BTN_1 5
2P_DOWN 6
2P_UP 7
2P_RIGHT 8
2P_LEFT 9
1 VCC

RA5
2.2k x 4
TILT 2
SERVICE 3
COIN_2 4
COIN_1 5
1 VCC

JAMMA_CONN
PWR_FLAG
GND 1
PWR_FLAG 2
VCC 3
4
5
6
+12V
COIN_CNT_1 8
COIN_LOCK_1 9
SPEAKER+ 10
JAMMA_11D 11
REDD 12
BLUED 13
GND 14
15
COIN_1 16
1P_START 17
1P_UP 18
1P_DOWN 19
1P_LEFT 20
1P_RIGHT 21
1P_BTN_1 22
1P_BTN_2 23
1P_BTN_3 24
25
26
27
GND 28
COIN_CNT_1 LOCK_COIL_1 SPEAKER_(+) NOT_USED RED BLUE VIDEO_GND TEST COIN_1 1P_START 1P_UP 1P_DOWN 1P_LEFT 1P_RIGHT 1P_BTN_1 1P_BTN_2 1P_BTN_3 1P_BTN_4 1P_SELECT GND GND
COIN_CNT_2 LOCK_COIL_2 SPEAKER_(-) NOT_USED GREEN SYNC SERVICE TILT COIN_2 2P_START 2P_UP 2P_DOWN 2P_LEFT 2P_RIGHT 2P_BTN_1 2P_BTN_2 2P_BTN_3 2P_BTN_4 2P_SELECT GND GND
A GND
B GND
C VCC
D
E
F
PWR_FLAG C1 470uF +12V
ZD1 > 5.0V
GND GND
J COIN_CNT_2
K COIN_LOCK_2
L
M SPEAKER-
N GREEN
P SYNC
R SERVICE
S TILT
T COIN_2
U 2P_START
V 2P_UP
W 2P_DOWN
X 2P_LEFT
Y 2P_RIGHT
Z 2P_BTN_1
a 2P_BTN_2
b 2P_BTN_3
c
d
e
f GND

+12V ZD2 > 12.0V C4 220uF GND GND

E12 X2-004
VCC 1
1P_START 2
1P_BTN_3 3
1P_BTN_2 4
1P_BTN_1 5
1P_DOWN 6
1P_UP 7
VCC 8
VCC1 VCC1 IN IN IN IN VCC2 VCC2 2k2 1k 2k2 1k
GND 16 OUT 15 1P_START_B OUT 14 1P_BTN_3_B OUT 13 1P_BTN_2_B OUT 12 1P_BTN_1_B OUT 11 1P_DOWN_B OUT 10 1P_UP_B GND 9 GND

D12 X2-004
VCC 1
1P_RIGHT 2
1P_LEFT 3
2P_START 4
2P_BTN_3 5
2P_BTN_2 6
2P_BTN_1 7
VCC 8
VCC1 VCC1 IN IN IN IN VCC2 VCC2 2k2 1k 2k2 1k
GND 16 OUT 15 1P_RIGHT_B OUT 14 1P_LEFT_B OUT 13 2P_START_B OUT 12 2P_BTN_3_B OUT 11 2P_BTN_2_B OUT 10 2P_BTN_1_B GND 9 GND

CC12 X2-004
VCC 1
2P_DOWN 2
2P_UP 3
2P_RIGHT 4
2P_LEFT 5
TILT 6
SERVICE 7
VCC 8
VCC1 VCC1 IN IN IN IN VCC2 VCC2 2k2 1k 2k2 1k
GND 16 OUT 15 2P_DOWN_B OUT 14 2P_UP_B OUT 13 2P_RIGHT_B OUT 12 2P_LEFT_B OUT 11 TILT_B OUT 10 SERVICE_B GND 9 GND

B12 X2-004
VCC 1
COIN_2 2
COIN_1 3
4
5
6
7
VCC 8
VCC1 VCC1 IN IN IN IN VCC2 VCC2 2k2 1k 2k2 1k
GND 16 OUT 15 COIN_2_B OUT 14 COIN_1_B OUT 13 OUT 12 OUT 11 OUT 10 GND 9 GND

RA2
8 x 10k
VCC 1
2 SWA1
3 SWA2
4 SWA3
5 SWA4
6 SWA5
7 SWA6
8 SWA7
9 SWA8

SWA
SW_DIP_x08
GND 16
15
14
13
12
11
10
9
1 SWA1
2 SWA2
3 SWA3
4 SWA4
5 SWA5
6 SWA6
7 SWA7
8 SWA8

SWA[1..8]

RA1
8 x 10k
VCC 1
2 SWB1
3 SWB2
4 SWB3
5 SWB4
6 SWB5
7 SWB6
8 SWB7
9 SWB8

SWB
SW_DIP_x08
GND 16
15
14
13
12
11
10
9
1 SWB1
2 SWB2
3 SWB3
4 SWB4
5 SWB5
6 SWB6
7 SWB7
8 SWB8

SWB[1..8]

H3 74HC257
SWA1 2 SWB1 3 SWA2 5 SWB2 6 SWA4 14 SWB4 13 SWA3 11 SWB3 10 SA0 1 15
IOa I1a IOb I1b IOc I1c IOd I1d S OE
Za 4 SD0
Zb 7 SD1
Zc 12 SD3
Zd 9 SD2

H2 74HC257
SWA5 2 SWB5 3 SWA6 5 SWB6 6 SWA8 14 SWB8 13 SWA7 11 SWB7 10 SA0 1 15
IOa I1a IOb I1b IOc I1c IOd I1d S OE
Za 4 SD4
Zb 7 SD5
Zc 12 SD7
Zd 9 SD6

F9C 74LS32
SA2 9
DIPSND 10
8

SA[0..2]
SD[0..7]
SD[0..7]

E10A 74LS139
GND 3
SA1 2
RDINPUTS 1
A1 A0 E decoder
O0 4
O1 5
O2 6
O3 7

D11 X1-004
SA0 37
E3 41
E2 40
E1
SD1 35
SD0 34
SD5 33
SD4 32
IOD1 IOD2 IOD3 IOD4
IOC1 18 TILT_B
IOC2 19 SERVICE_B
IOC3 20 COIN_2_B
IOC4 21 COIN_1_B
1P_START_B 2
1P_BTN_3_B 3
1P_BTN_2_B 4
1P_BTN_1_B 5
1P_DOWN_B 6
1P_UP_B 7
1P_RIGHT_B 8
1P_LEFT_B 9
INA1 INA2 INA3 INA4 INA5 INA6 INA7 INA8
D7 51 SD7
D6 50 SD6
D5 49 SD5
D4 48 SD4
D3 47 SD3
D2 46 SD2
D1 45 SD1
D0 44 SD0
2P_START_B 10
2P_BTN_3_B 11
2P_BTN_2_B 12
2P_BTN_1_B 13
2P_DOWN_B 14
2P_UP_B 15
2P_RIGHT_B 16
2P_LEFT_B 17
INB1 INB2 INB3 INB4 INB5 INB6 INB7 INB8
COIN_LOCK_1_B 22
COIN_LOCK_2_B 23
COIN_CNT_2_B 24
COIN_CNT_1_B 25
IOA1 IOA2 IOA3 IOA4
IOB1 31 (SBSWD5)
IOB2 30 (SBSWD5)
IOB3 29 SBSWD3
IOB4 28 SBSWD2
38
36
43
42
39
UK UK UK UK UK
VCC 52 VCC
VCC 27
GND 26 GND
1
GND VCC GND

SBSWD4 JP7 SBSWD5
SBSWD[2..5]

F12
COIN_LOCK_2_B 3
COIN_CNT_2_B 6
COIN_CNT_1_B 11
COIN_LOCK_1_B 14
I1 I2 I3 I4 TD62064
O1 2 COIN_LOCK_2
O2 7 COIN_CNT_2
O3 9 COIN_CNT_1
O4 16 COIN_LOCK_1
+12V 1
8
10
15
COM GND COM GND NC GND NC GND
4
5
12
13
GND

The I/O direction of each port in X1-004 is probably determined by connecting the pins 36,38,39, 42 and 43 to VCC and GND accordingly.

In tnzs IOD -> IOC, IOB -> IOA

E1	E2	E3	Active output
H	L	L	INA-> D, IOB -> IOA
H	L	H	INB -> D, IOB -> IOA
L	H	L	IOC -> IOD, IOB -> IOA
X	X	X	IOB -> IOA

All inputs are active low, when the outputs are not active they are in tri-state state. Except for IOA in this case.

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Sheet: /IO/
File: io.kicad_sch
Title: The NewZealand Story - PO-043A
Size: A3 | Date: 2024-04-07 | Rev:
KiCad E.D.A. 8.0.5 | Id: 6/7

YMROMA[14..15]

S2A[14..15]

S2A[0..15]

DA[0..7]

S2D[0..7]

0x0000..0x7FFF ROML
0x8000..0xBFFF S2ROMHCS
0xC000..0xDFFF S2RAMCS

S2ROMHCS = S2A15 & S2A14 & S2MREQ
ROMCS = (S2A15 & S2A14 + S2A15) S2MREQ

F7
74LS157
S Quad Mux

B53-26.U34
27512

J6
27C010
Not mounted on tnzs

J9
HA17408

J8
YM2203

H6A
74LS139
decoder

U34
Z80CPU

J5
HM65256BLP-12

H6B
74LS139
decoder

H4
74HC374
3-state flip flops

F8A
74LS74

F9D
74LS32

F8B
74LS74

74LS04

HH9
YM3014

RC4558

MB3730
INV. AMP

JP10
JP8
JP9

JAMMA_11

SPEAKER+
SPEAKER-

+5V
-5V
+12V
VCC
GND

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Sheet: /Sound/
File: sound.kicad_sch

Title: The NewZealand Story – P0-043A

Size: A3	Date: 2024-04-07	Rev:
KiCad E.D.A. 8.0.5		Id: 7/7